

Отчёт по лабораторной работе 8

Адресация IPv4 и IPv6. Настройка маршрутизации

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Цели и задачи работы

Цель лабораторной работы

Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования в среде GNS3.

Выполнение лабораторной работы

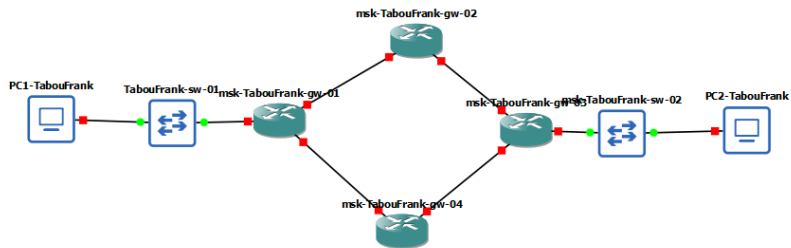
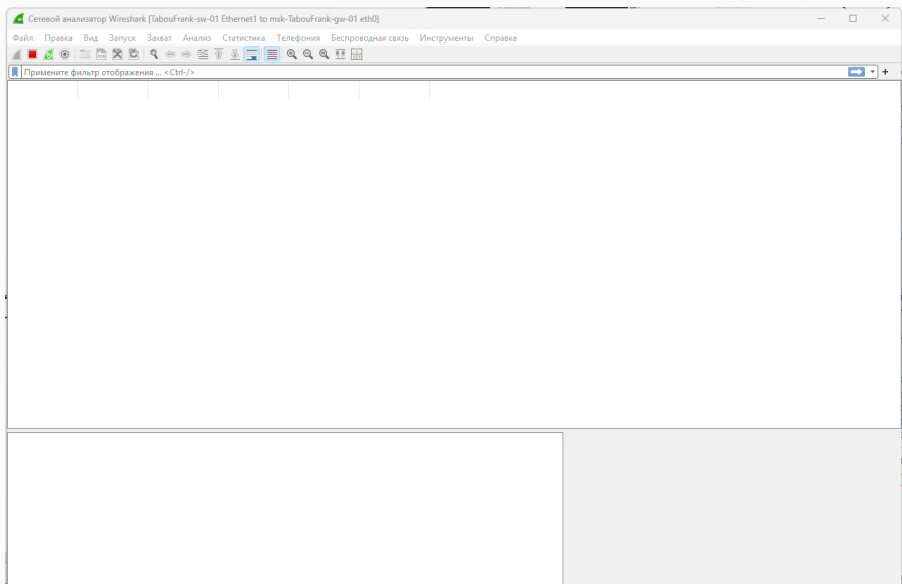
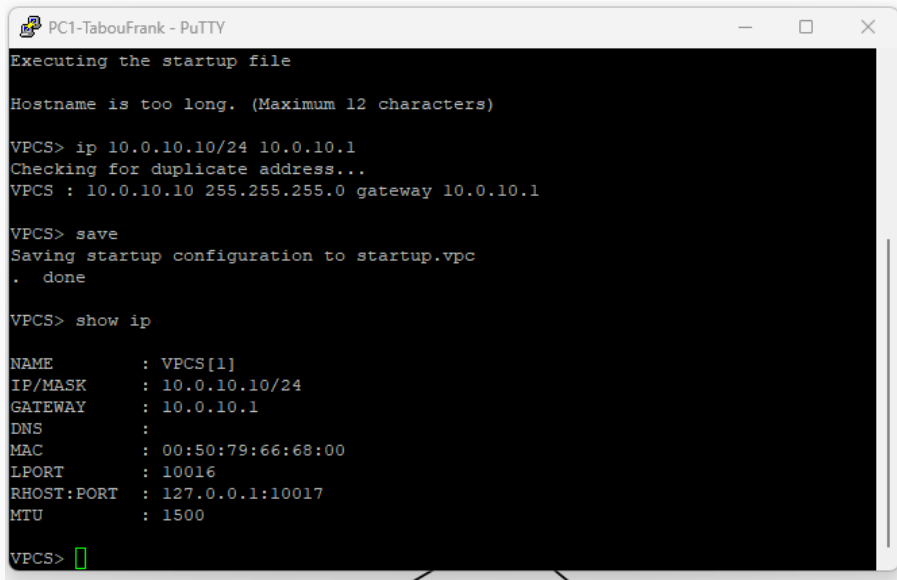


Рис. 1: Топология сети в GNS3

Настройка IPv6 на PC1



Настройка IPv6 на PC2



```
PC1-TabouFrank - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

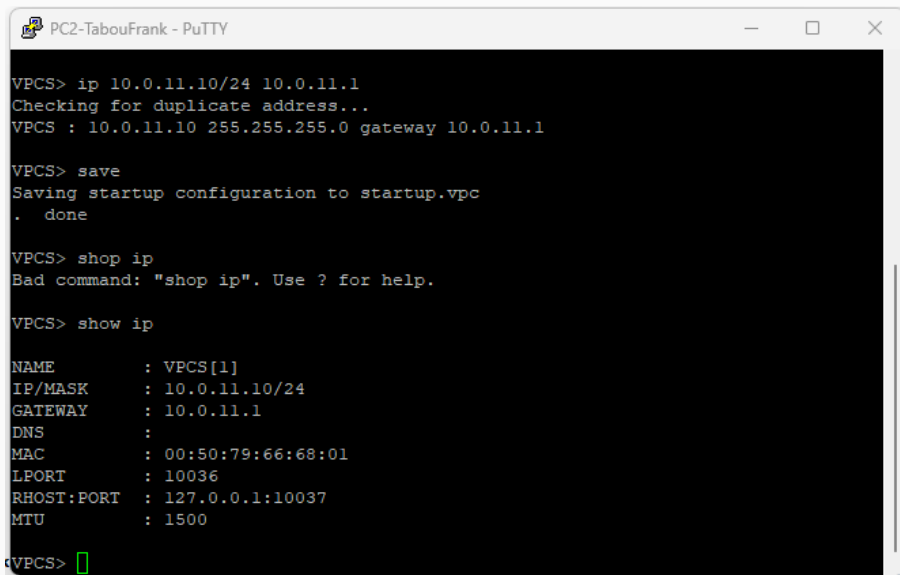
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 10.0.10.10/24
GATEWAY     : 10.0.10.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 10016
RHOST:PORT  : 127.0.0.1:10017
MTU         : 1500

VPCS> 
```

Подготовка gw-01



```
PC2-TabouFrank - PuTTY

VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> shop ip
Bad command: "shop ip". Use ? for help.

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.11.10/24
GATEWAY    : 10.0.11.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 10036
RHOST:PORT : 127.0.0.1:10037
MTU        : 1500

VPCS> 
```


Адресация gw-01

```
msk-frank-gw-01 ~ #
* Setting keymap ... [ ok ]
* Starting networking ... *   lo ... [ ok ]
* Starting busybox syslog ... [ ok ]
* Seeding 256 bits and crediting
* Saving 256 bits of creditable seed for next boot
* Starting busybox acpid ... [ ok ]
* Starting busybox crond ... [ ok ]
Started watchfrr
* Starting sshd ... [ ok ]

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-frank-gw-01
msk-frank-gw-01(config)# interface eth0
msk-frank-gw-01(config-if)# ip address 10.0.10.1/24
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# interface eth1
msk-frank-gw-01(config-if)# ip address 10.0.1.1/24
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)#
```

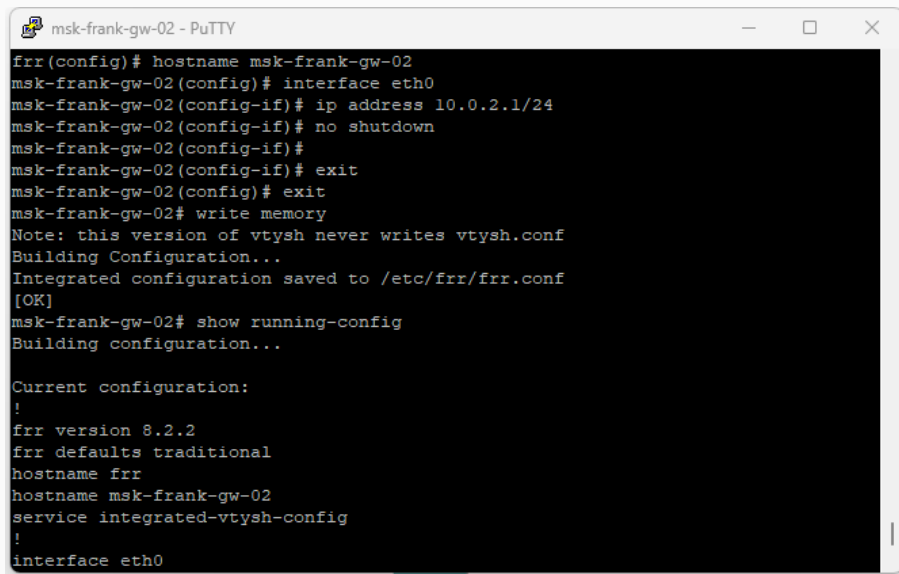
Подготовка gw-02

```
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-frank-gw-01#
```

Рис. 6: Подготовка gw-02

Адресация gw-02



```
msk-frank-gw-02 - PuTTY

frr(config)# hostname msk-frank-gw-02
msk-frank-gw-02(config)# interface eth0
msk-frank-gw-02(config-if)# ip address 10.0.2.1/24
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)#
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-02
service integrated-vtysh-config
!
interface eth0
```

Подготовка и адресация gw-03

```
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth2
msk-frank-gw-03(config-if)# ip address 10.0.3.1/24
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# exit
msk-frank-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-03
service integrated-vtysh-config
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
interface eth0
 ip address 10.0.11.1/24
```

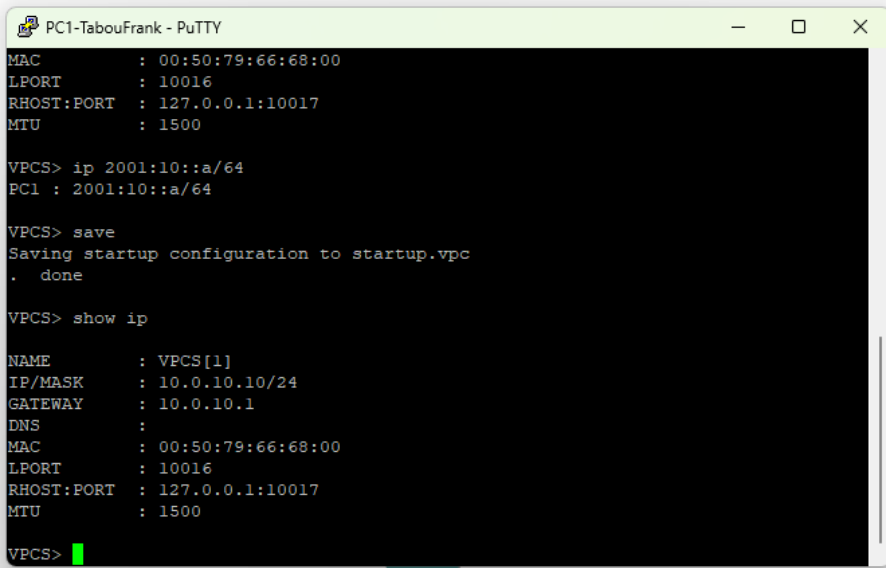
Подготовка и адресация gw-03

```
msk-frank-gw-04 - PuTTY
Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiko Ishiguro, et al.

frr# configure terminal
frr(config)# hostname frank-gw-04
% Unknown command: hostname frank-gw-04
frr(config)# hostname msk-frank-gw-04
msk-frank-gw-04(config)# interface eth0
msk-frank-gw-04(config-if)# ip address 10.0.3.2/24
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# configure terminal
msk-frank-gw-04(config)# interface eth1
msk-frank-gw-04(config-if)# ip address 10.0.4.1/24
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-frank-gw-04#
```

Проверка RA на оконечных устройствах



```
PC1-TabouFrank - PuTTY
MAC      : 00:50:79:66:68:00
LPORT    : 10016
RHOST:PORT : 127.0.0.1:10017
MTU      : 1500

VPCS> ip 2001:10::a/64
PC1 : 2001:10::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME      : VPCS[1]
IP/MASK   : 10.0.10.10/24
GATEWAY   : 10.0.10.1
DNS       :
MAC       : 00:50:79:66:68:00
LPORT     : 10016
RHOST:PORT : 127.0.0.1:10017
MTU       : 1500

VPCS> 
```

Проверка RA на оконечных устройствах

```
PC2-TabouFrank - PuTTY
LPORT      : 10036
RHOST:PORT  : 127.0.0.1:10037
MTU         : 1500

VPCS> ip 2001:11::a/64
PC1 : 2001:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME          : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE    : 2001:11::a/64
DNS            :
ROUTER LINK-LAYER :
MAC           : 00:50:79:66:68:01
LPORT         : 10036
RHOST:PORT     : 127.0.0.1:10037
MTU           : 1500

VPCS> █
```

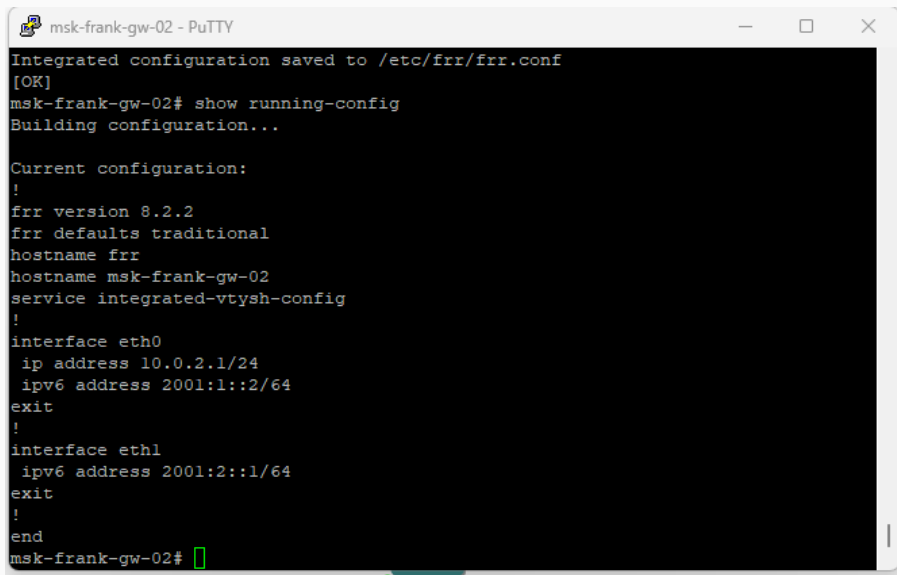
Ping с gw-01 до настройки маршрутизации

```
msk-frank-gw-01 - PuTTY
ip address 10.0.4.2/24
exit
!
end
msk-frank-gw-01# configure terminal
msk-frank-gw-01(config)# ipv6 forwarding
msk-frank-gw-01(config)# interface eth0
msk-frank-gw-01(config-if)# ipv6
% Command incomplete: ipv6
msk-frank-gw-01(config-if)# ipv6 address 2001:10::1/64
msk-frank-gw-01(config-if)# no ipv6 nd suppress-ra
msk-frank-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# interface eth1
msk-frank-gw-01(config-if)# ipv6 address 2001:1::1/64
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# interface eth2
msk-frank-gw-01(config-if)# ipv6 address 2001:4::2/64
msk-frank-gw-01(config-if)# no shutdown
msk-frank-gw-01(config-if)# exit
msk-frank-gw-01(config)# exit
msk-frank-gw-01#
```


Анализ трафика до RIP (gw-01 ↔ gw-03)

```
msk-frank-gw-02 - PuTTY
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.2.1/24
exit
!
end
msk-frank-gw-02# configure terminal
msk-frank-gw-02(config)# ipv6 forwarding
msk-frank-gw-02(config)# interface eth0
msk-frank-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# interface eth1
msk-frank-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-frank-gw-02(config-if)# no shutdown
msk-frank-gw-02(config-if)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02#
```

Настройка RIP на gw-01



```
msk-frank-gw-02 - PuTTY
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.2.1/24
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ipv6 address 2001:2::1/64
exit
!
end
msk-frank-gw-02#
```

Настройка RIP на gw-02

```
msk-frank-gw-03 - PuTTY
interface eth0
 ip address 10.0.11.1/24
exit
!
end
msk-frank-gw-03# configure interface
% Unknown command: configure interface
msk-frank-gw-03# configure terminal
msk-frank-gw-03(config)# ipv6 forwarding
msk-frank-gw-03(config)# interface eth0
msk-frank-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-frank-gw-03(config-if)# no ipv6 nd suppress-ra
% Unknown command: no ipv6 nd suppress-ra
msk-frank-gw-03(config-if)# no ipv6 nd suppress-ra
msk-frank-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth1
msk-frank-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# interface eth2
msk-frank-gw-03(config-if)# ipv6 address 2001:3::1/64
msk-frank-gw-03(config-if)# no shutdown
msk-frank-gw-03(config-if)# exit
msk-frank-gw-03(config)# exit
msk-frank-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03#
```

Настройка RIP на gw-03

```
msk-frank-gw-03 - PuTTY
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-03# show running-config
Building configuration...

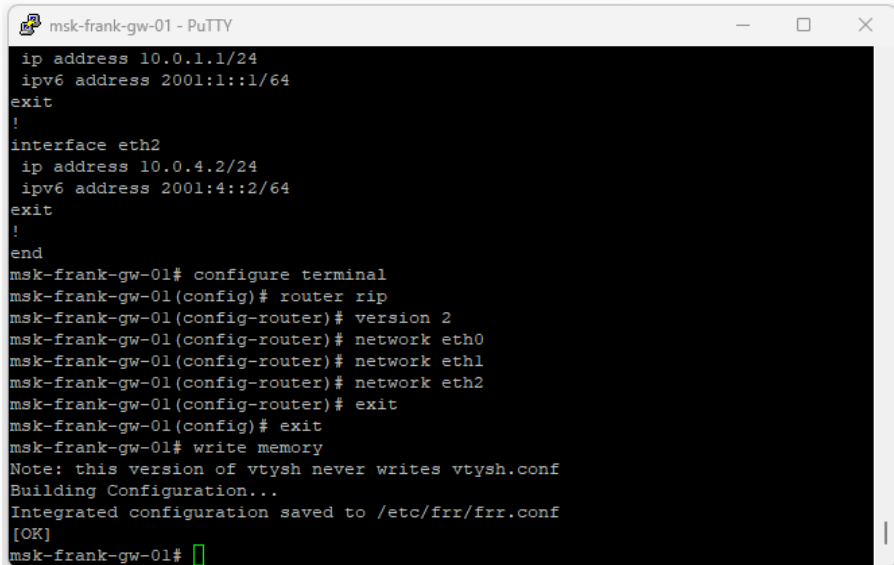
Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-03
service integrated-vtysh-config
!
interface eth0
    ipv6 address 2001:11::1/64
    ipv6 nd prefix 2001:11::/64
    no ipv6 nd suppress-ra
exit
!
interface eth1
    ip address 10.0.2.2/24
    ipv6 address 2001:2::2/64
exit
!
interface eth2
    ip address 10.0.3.1/24
    ipv6 address 2001:3::1/64
exit
!
interface etho
lines 1-23
```

Проверка связности после включения RIP

```
msk-frank-gw-04 - PuTTY
ip address 10.0.4.1/24
exit
!
end
msk-frank-gw-04# configure terminal
msk-frank-gw-04(config)# ipv6 forwarding
msk-frank-gw-04(config)# ipv6 address 2001:3::2/64
% Unknown command: ipv6 address 2001:3::2/64
msk-frank-gw-04(config)# interface eth0
msk-frank-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# interface eth1
msk-frank-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04# show running-config
Building configuration...

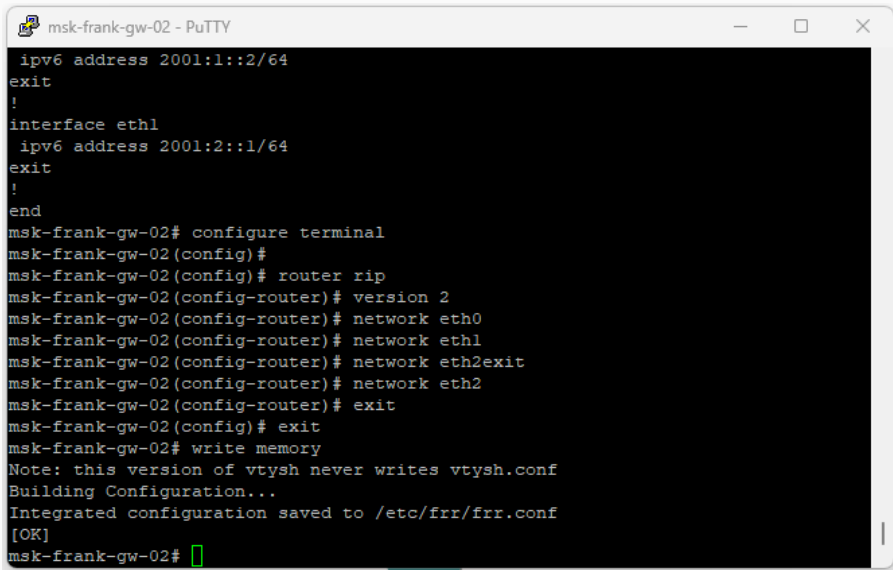
Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
 exit
!
interface eth1
 ip address 10.0.4.1/24
 ipv6 address 2001:4::1/64
 exit
!
end
msk-frank-gw-04#
```

Анализ трафика после RIP



```
msk-frank-gw-01 - PuTTY
ip address 10.0.1.1/24
ipv6 address 2001:1::1/64
exit
!
interface eth2
ip address 10.0.4.2/24
ipv6 address 2001:4::2/64
exit
!
end
msk-frank-gw-01# configure terminal
msk-frank-gw-01(config)# router rip
msk-frank-gw-01(config-router)# version 2
msk-frank-gw-01(config-router)# network eth0
msk-frank-gw-01(config-router)# network eth1
msk-frank-gw-01(config-router)# network eth2
msk-frank-gw-01(config-router)# exit
msk-frank-gw-01(config)# exit
msk-frank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-01#
```

Туннель SIT на gw-01



```
msk-frank-gw-02 - PuTTY
ipv6 address 2001:1::2/64
exit
!
interface eth1
  ipv6 address 2001:2::1/64
exit
!
end
msk-frank-gw-02# configure terminal
msk-frank-gw-02(config)#
msk-frank-gw-02(config)# router rip
msk-frank-gw-02(config-router)# version 2
msk-frank-gw-02(config-router)# network eth0
msk-frank-gw-02(config-router)# network eth1
msk-frank-gw-02(config-router)# network eth2exit
msk-frank-gw-02(config-router)# network eth2
msk-frank-gw-02(config-router)# exit
msk-frank-gw-02(config)# exit
msk-frank-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-02#
```

Туннель SIT на gw-02

```
msk-frank-gw-04 - PuTTY
msk-frank-gw-04(config)# interface eth1
msk-frank-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-frank-gw-04(config-if)# no shutdown
msk-frank-gw-04(config-if)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-frank-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
 exit
!
interface eth1
 ip address 10.0.4.1/24
 ipv6 address 2001:4::1/64
 exit
!
end
msk-frank-gw-04# configure terminal
msk-frank-gw-04(config)# router rip
msk-frank-gw-04(config-router)# version 2
msk-frank-gw-04(config-router)# network eth0
msk-frank-gw-04(config-router)# network eth1
msk-frank-gw-04(config-router)# exit
msk-frank-gw-04(config)# exit
msk-frank-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-frank-gw-04#
```


Статическая маршрутизация IPv6

```
msk-frank-gw-04# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 12 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2     2
    eth1           2     2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway         BadPackets  BadRoutes   Distance  Last Update
    10.0.3.1         0           0           120      00:00:08
    10.0.4.2         0           0           120      00:00:08
  Distance: (default is 120)
msk-frank-gw-04#
```

Рис. 21: Статический IPv6-маршрут на gw-01

Статическая маршрутизация IPv6

```
msk-frank-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/3] via 10.0.4.1, eth2, weight 1, 00:04:38
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:04:38
msk-frank-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

        Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24         0.0.0.0           1 self              0
R(n) 10.0.2.0/24         10.0.4.1          3 10.0.4.1           0 02:42
R(n) 10.0.3.0/24         10.0.4.1          2 10.0.4.1           0 02:42
C(i) 10.0.4.0/24         0.0.0.0           1 self              0
C(i) 10.0.10.0/24        0.0.0.0           1 self              0
msk-frank-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 27 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface    Send  Recv  Key-chain
    eth0         2     2
    eth1         2     2
    eth2         2     2
  Routing for Networks:
    eth0
```

Проверка с PC1: ping и trace до PC2

```
PC1-TabouFrank - PuTTY
MTU           : 1500

VPCS> ping 10.0.11.10

*10.0.10.1 icmp_seq=1 ttl=64 time=1.073 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=2 ttl=64 time=1.484 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=3 ttl=64 time=2.758 ms (ICMP type:3, code:0, Destination net
work unreachable)
*10.0.10.1 icmp_seq=4 ttl=64 time=3.314 ms (ICMP type:3, code:0, Destination net
work unreachable)
10.0.11.10 icmp_seq=5 timeout

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  *10.0.10.1  1.450 ms (ICMP type:3, code:0, Destination network unreachable
)  *
 2      * * *
 3      * * *
 4      * * *
 5      * * *
 6      * * *
```

Проверка с PC2: ping e trace до PC1

```
msh-frank-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.4.1          3 10.0.4.1          0 02:35
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1          0 02:35
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
msh-frank-gw-01#
```

Рис. 24: PC2 → PC1: ping/trace

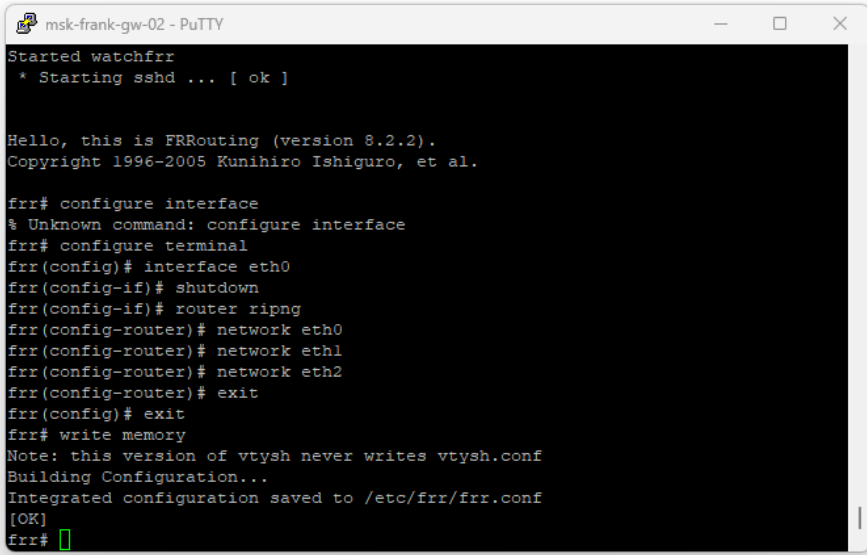
Трафик IPv6 при передаче через IPv4

```
msk-frank-gw-01 - PuTTY
* Saving 256 bits of creditable seed for next boot
* Starting busybox acpid ... [ ok ]
* Starting busybox crond ... [ ok ]
Started watchfrr
* Starting sshd ... [ ok ]

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# show ip rip
% Unknown command: show ip rip
frr(config)# router ripng
frr(config-router)# network eth0
frr(config-router)# network eth1
frr(config-router)# exit
frr(config)# exit
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr#
```

Где видна информация о прохождении через туннель



```
msk-frank-gw-02 - PuTTY
Started watchfrr
* Starting sshd ... [ ok ]

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure interface
% Unknown command: configure interface
frr# configure terminal
frr(config)# interface eth0
frr(config-if)# shutdown
frr(config-if)# router ripng
frr(config-router)# network eth0
frr(config-router)# network eth1
frr(config-router)# network eth2
frr(config-router)# exit
frr(config)# exit
frr# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
frr#
```

Выводы по trabalho

В GNS3 настроена сеть с VyOS и VPCS, выполнена адресация IPv4/IPv6 и проведена проверка связности. Динамическая маршрутизация IPv4 (RIP) обеспечила доступность удалённых IPv4-сетей. Туннель SIT и статические маршруты IPv6 обеспечили обмен трафиком между удалёнными IPv6-сегментами. Анализ Wireshark подтвердил обмен ARP/ICMP/RIP и наличие инкапсуляции IPv6-пакетов внутри IPv4 при передаче через туннель.